Compact Unpaired Style Transfer for a Disposable-Camera Aesthetic, with Synthesised Source Domains and Held-Out Model Selection

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Abstract

Learning a photographic aesthetic from a target domain alone is the binding constraint of unpaired image-to-image translation, and it is usually left unsolved. We present NIGHT FLASH EDGE, a 4.02 M-parameter generator that learns a disposable-camera / Night Flash aesthetic for a native mobile camera, subject to two hard requirements: real-time execution on device, and export to Core ML.

We make three contributions. First, we identify and correct a systematic bias in the standard unpaired objective: when an L_1 content penalty is evaluated on a *stochastic* output, it penalises the stochastic texture the model is being asked to produce. We measure the penalty at 1.9× the magnitude of the entire learned residual, and show that separating the L_1 and edge halves across different images removes it. Second, we introduce a synthetic degradation pipeline that manufactures a source domain from the target photographs by simulating a non-flash camera, in two variants: one with exact correspondence, which admits supervised reconstruction and real quality metrics, and one without, which provably removes the inverse-mapping shortcut the first variant admits. Third, we supply a validation regime that makes model selection possible at all, using reproducible held-out strips, a per-epoch metric history, and best-checkpoint retention — none of which a loss curve provides.

We report measurements for every claim, including two places where measurement falsified our initial design claims, and report the resulting corrections.

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1 Introduction

A mobile camera application typically ships a deterministic colour transform plus a finishing chain, and can render a fixed aesthetic reliably. What it cannot do is *adapt*: choose how much flash falloff a given scene wants, how strongly to lift a shadow, where grain should sit. A learned filter can, provided it is small enough to run in real time and simple enough to export to a hardware accelerator.

The obstacle is data. A photographic aesthetic is a property of lighting and film response, not of content, so the pairs one would want — the same scene with and without the treatment — do not exist. Training must therefore be unpaired, and unpaired training brings a specific difficulty that this work addresses directly: *with only a target domain, there is no source domain to translate from.*

Three questions structure the paper. §3 asks whether the standard unpaired objective is correctly specified when the generator emits stochastic texture, and finds that it is not. §4 asks how a source domain can be manufactured when only a target exists. §5 asks how model selection is possible without paired validation data, which is the precondition for knowing whether any of the above worked.

Contributions.

1. A measurement and correction of texture bias in the unpaired content penalty (§3).
2. A nine-stage camera-degradation model for synthesising a source domain, with a paired and an unpaired pairing scheme, and an analysis of what it can and cannot achieve (§4).
3. A held-out validation regime with deterministic reproduction, per-image comparability, a metric history, and best-checkpoint selection (§5).
4. A released, tested implementation in which all 28 architecture × head combinations train, infer and convert to Core ML (§7).

2 Method

2.1 Model

The generator predicts a bounded residual on the input:

$$x_{\text{out}} = \text{clamp}(x_{\text{in}} + \Delta_\theta(x_{\text{in}}), 0, 1)\,, \qquad \|\Delta_\theta\|_\infty \le \rho. \tag{1}$$

Eq. (1) is the load-bearing design decision. The only way to produce a pixel is to perturb a pixel that already exists, so faces, text and geometry are structurally protected rather than protected by a penalty term. It also fixes the failure direction: an under-trained model degrades to a near-identity filter, which is the correct behaviour for a camera filter.

The trunk is a convolutional encoder/decoder with local-window self-attention (`EdgeViT`) and a single overlapping cross-attention stage (`HAT`) for global context, at 4.02 M parameters. Three photometric capabilities — halation, a subject gate, and per-region exposure — are each gated by a zero-initialised mix scalar, so a fresh model is exactly the plain residual and each capability engages only once training motivates it.

The head determines the permitted parameterisation. We use a frequency-split head by default, predicting low, mid and high bands separately, on the grounds that grain and halation are high-frequency phenomena that a per-pixel head represents poorly.

2.2 Objective

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda_c \mathcal{L}_{\text{content}} + \lambda_m \mathcal{L}_{\text{moment}} + \lambda_i \mathcal{L}_{\text{identity}} + \lambda_e \mathcal{L}_{\text{effect}}, \tag{2}$$

with $(\lambda_c, \lambda_m, \lambda_i, \lambda_e) = (1.5, 0.35, 0.75, 0.02)$ and a PatchGAN discriminator at unit weight. The discriminator is the only term that sees targets as targets; every other term is a constraint on the generator.

3 A correction: texture bias in the content penalty

3.1 The observation

The generator emits a deterministic residual, which a seeded procedural stage then turns into a stochastic image by adding grain, leak, exposure, saturation and vignette. Let x_{det} denote the pre-effect output and x_{styled} the post-effect output.

The content term is a convex combination of an L_1 half and a Sobel edge half. Both halves were originally evaluated on x_{styled}. Because L_1 is texture sensitive, this charges the output for the stochastic texture.

3.2 Measurement

We train a model on a small pool and measure both penalties directly. The predicate "L_1 on the styled image" and "L_1 on the deterministic image" differ by construction, but the interesting quantity is the magnitude:

Quantity	Value
$\text{content}(x_{\text{det}}, x_{\text{src}})$	0.01451
$\text{content}(x_{\text{styled}}, x_{\text{src}})$	0.04237
Penalty incurred by the effects	+0.02786
$\lVert x_{\text{det}} - x_{\text{src}} \rVert_1$	0.00891
$\lVert x_{\text{styled}} - x_{\text{det}} \rVert_1$	0.01601

Table 1: The stochastic effects are charged roughly 1.9× more by the content term than the entire learned residual is charged for the whole style transfer.

Table 1 is the whole argument. The adversarial term, at unit weight, asks for grain. The content term, at weight 1.5, penalises it, by an amount nearly double to the residual's own contribution. The content term wins, and the model learns to suppress the very effects its own effect head exists to produce. This is the familiar failure where a squared or absolute error on a stochastic target averages the stochastic component away.

3.3 The correction

The two halves belong to different comparisons. L_1 is a texture statistic and must be evaluated where the texture is not stochastic; the Sobel edge response is a structure statistic, is indifferent to fine texture, and should be evaluated on the final image so that broken geometry is actually caught:

$$\mathcal{L}_{\text{content}} = 0.65\, L_1(x_{\text{det}},\, x_{\text{src}}) + 0.35\, \mathcal{E}\big(x_{\text{styled}},\, x_{\text{src}}\big)\,. \tag{3}$$

Eq. (3) leaves the intent of the original term — stay close to the source — intact while removing the gradient that fights the texture.

Consequence for comparability. Grain and halation become visibly stronger, which is the intent, but any checkpoint trained under the previous balance is no longer comparable. We

therefore version the objective (`LOSS_VERSION`= 2), record the version in every checkpoint and in `config.json`, and warn on resume when it differs, so that the two halves of a run are never silently pooled.

3.4 Related failure avoided in the supervised term

Where a correspondence exists, we add a reconstruction term. It is deliberately evaluated on x_{det} for the same reason: an L_1 term pointed at a stochastic *target* would average the target's own grain out of both sides.

4 Synthesised source domains

4.1 Motivation

Unpaired translation assumes a source domain. When only a target domain exists, one must be constructed, or the problem is not posed. The construction must be a plausible capture of the same scenes by a different camera, so that the learned map generalises to real photographs rather than to the construction's fingerprints.

4.2 Pipeline

We simulate a non-flash camera by undoing a capture pipeline, resampling first so that photometric operations act on clean pixels:

#	Stage
1	crop and small rotation, shared with the target
2	downscale then upscale — removes grain and fine detail
3	Gaussian blur — residual softening
4	contrast compression about mid-grey
5	black lift, $x \mapsto x + b(1 - x)$, preserving white
6	exposure gain
7	white-balance shift, warmed
8	saturation increase
9	light sensor noise

Stages 4–5 carry most of the shift. Compressing the tonal range is what removes the flash falloff signature — the bright subject on dark ground — and it is what makes the output read as an evenly-lit non-flash capture rather than merely a dark one.

4.3 Effect, measured

Averaged over eight draws on a real photograph:

Statistic	`subtle`	`moderate`	`strong`
high-frequency energy (relative)	0.52	0.30	0.21
tonal range $p_{99} - p_1$	0.75	0.57	0.49
warmth $R - B$	0.040	0.078	0.121

4.4 Two claims we had to retract

Resampling does not remove halation. We initially documented the downscale/upscale stage as destroying halation bloom. A targeted probe refuted this. Halation is a wide, low-frequency phenomenon; a low-pass filter has nothing to remove, and blurring a glow makes it *brighter* by spreading the core into the surrounding ring. On a probe with a specular point and a halo, the ring-to-peak luminance ratio rises from 0.57 to 0.86 as blur increases from 0 to 4. The

pipeline attenuates halation, driven by tonal compression rather than filtering, which suffices for our purpose — the model must learn to add glow where the target has it — but would not suffice for constructing a source with provably zero halation. That would require re-rendering the illumination.

Too-strong degradation is a defect. A first `strong` preset used blur to 2.5 and black lift to 0.20. The output looked foggy rather than like a photograph, and its injected noise was conspicuous against the softened background. Training on such inputs teaches a model to expect a distribution that never occurs at inference, which is a worse outcome than a weaker domain shift. We pulled the ranges back until the strongest profile still reads as a believable overcast capture. Plausibility, not magnitude, is the design objective.

4.5 Two pairing schemes

Scheme	Source	Target	Correspondence
`paired`	$\deg(x_i)$	x_i	exact, same file
`sameset`	$\deg(x_j),\ j \neq i$	x_i	none, one pool

In both, the shared geometric warp is applied to both images first and the degradation runs afterwards on the source only, so any crop, flip or rotation is common to the pair and only the simulated camera differs.

4.6 The inverse-mapping shortcut

The `paired` scheme admits a shortcut. Because the source is a deterministic function of the target, the network can reduce its loss by learning $\deg^{-1}$ rather than the aesthetic. It will score well on freshly-degraded inputs and fail on real photographs, and the failure is invisible unless sought.

The `sameset` scheme removes the shortcut by construction: x_i and x_j are unrelated photographs, so no pointwise relationship exists to invert and the model must learn the conditional distribution of the look. It also doubles the usable pairings, since every image serves as both source and target.

The price is that without correspondence no supervised reconstruction is available and no real quality number can be computed, so only distribution-level metrics apply. We accept this asymmetry rather than pretending it away, and the practical recommendation is to run both schemes as parallel arms and let the evidence decide.

4.7 Variation across images, epochs and runs

Parameters are drawn per image; the dataset seeds its generator with the epoch index, so the draw also varies per epoch at no cost; and the run seed shifts the stream. Two further mechanisms are provided. An *identity path*, taken with probability 0.10, feeds the target through undegraded, anchoring the identity case and widening the input distribution. A *curriculum*, on by default, ramps the profile `subtle` → `moderate` → `strong`, giving early epochs an easy mapping and later epochs robustness.

5 Held-out validation and model selection

5.1 The problem

A discriminator loss is not a quality signal, and neither is a generator loss. Without a validation signal there is no basis for choosing between checkpoints, no basis for detecting overfitting, and no basis for deciding whether a hyperparameter change helped. In this setting the difficulty is sharper still, because the data is unpaired by construction.

5.2 Deterministic reproduction

Two independent properties must be pinned, and the second is unconditional.

Which images appear is set by `val_select`; we default to `fixed`, because per-image progress is only legible if the same images recur, and because re-showing the same handful would otherwise hide a model that has memorised them.

How each image is degraded and seeded is always pinned, with seeds derived from the filename by SHA-256 rather than from the epoch-dependent random stream. This is not an optimisation. A validation image degraded with a fresh draw each epoch is a different problem each time, and any apparent improvement is noise. Python's built-in `hash()` is salted per interpreter and is therefore unusable for this purpose; cross-session stability requires a cryptographic digest.

The resulting artefact is one strip image per epoch: one image per row, cells ordered input, output, target, so a given image's input and output are adjacent. The header carries the epoch, mode, active profile and aggregate metrics.

5.3 From measurement to selection

The strip is for a person; the CSV history is for the log. One row per epoch of PSNR, SSIM, L_1 and effect magnitude makes the curve plottable, and — more importantly — makes checkpoint selection possible. Before this, only `latest.pt` existed: a run peaking at epoch 40 and decaying by epoch 60 had no way to recover its best weights. We now retain a `best.pt` on improvement and prune old per-epoch checkpoints, since 60 epochs of a 17 MB checkpoint is a gigabyte that also slows every snapshot.

We write `best.pt` only in the `paired` scheme. In `sameset`, PSNR compares two unrelated photographs, so ranking on it would be meaningless, and we decline to emit a checkpoint that appears authoritative but is not.

5.4 A shared-warp defect

The first implementation of the shared geometric warp sampled scale, rotation and flips once but drew the crop *offset* from the shared random stream inside each application, so the source and target received different offsets. This is harmless for unpaired data, where the images are unrelated anyway, and it silently destroys the correspondence for paired data. We sample the offset once as a fraction and clamp it per image, so differently sized images still receive the same relative crop. We report it because the class of bug — a correspondence silently destroyed by a shared mutable stream — is easy to introduce and invisible without a test that asserts byte-identical outputs from identical inputs.

6 Deployment

6.1 Inference

Whole-frame inference is the default: the image is letterboxed, styled, and restored to its exact original dimensions, with no cropping and no stretching. We verify a landmark round trip at 0.0017 worst positional error with output size identical to input.

Tiled inference, for native-resolution detail, crossfades overlapping tiles under a raised-cosine window. The tile size must match the training crop; a mismatch is a distribution shift, and the implementation warns rather than blocks. Measured cost on a 4032×3024 source at 256 px tiles is 336 tiles and 22.0 MP at 25% overlap, rising to 713 tiles and 46.7 MP at 50%. The relevant quantity is the overlap *ratio*, not its absolute value.

Frame borders do not fade, which is the usual failure of naive windowed blending. The window carries a positive floor and edge tiles extend past the frame, so the frame border is never a tile border; a 255-valued border strip survives at 254.6/255 against a worst-case error of 1.0.

6.2 Core ML

All 28 architecture × head combinations convert, with tensor-I/O mean absolute error 0.00030. Several choices exist solely to satisfy the export boundary and should not be refactored away: explicit attention matmuls in place of the fused multi-head operator, a flat $(s^3, 3)$ LUT table, explicit trilinear gathers in place of `grid_sample`, and shape integers precomputed at construction because `torch.jit.trace` turns `x.shape[1]` into a zero-dimensional tensor that breaks pooling and `min`/`max`.

6.3 Training/inference gap

Synthesised source domains narrow the input distribution relative to arbitrary user photographs. We mitigate this on the training side — wide randomisation, the identity path, and a strong profile — and we explicitly reject inference-time input normalisation, which risks double-degrading a user image.

7 Implementation and verification

264 tests pass. Beyond shape and conversion checks, the suite pins the behaviours that would otherwise fail silently: the content-loss split, including that the L_1 half is insensitive to the styled image and the edge half is not; that a shared warp yields byte-identical source and target from identical inputs; that validation degradation is invariant across epochs while training degradation is not; that the `sameset` scheme never pairs an image with itself and degrades gracefully on a single-image pool; that `best.pt` is not written without a correspondence; and an abstract-syntax check that no module unpacks a generator call without the architecture-agnostic helper, a defect that had already shipped once in the evaluation entry point.

8 Conclusion

The most consequential finding here is unglamorous: a standard loss term, applied in a plausible place, silently cancelled the component of the model it was supposed to support. It was found only by measuring the penalty each stage incurits rather than by inspecting the loss curve, which continued to fall. The second is that a validation regime is a prerequisite for knowing whether anything worked, and that without a correspondence it must be built from distributional evidence instead of reconstruction metrics. Both are, we think, general lessons for small unpaired systems rather than facts about this particular aesthetic.

Limitations. The degradation model attenuates rather than eliminates halation, and the strongest profile is a judgement call about plausibility. The Night Flash Edge aesthetic is narrow by construction; multiple distinct looks would require separate checkpoints or a style embedding. Unpaired training cannot guarantee identity preservation, and the held-out metrics reported here are quality proxies rather than perceptual ground truth.

A Reproduction

```
# train, paired scheme, strip and best.pt every epoch
nf train --target-dir POOL --name a --epochs 60 --val-count 8

# the same pool with the inverse-mapping shortcut removed
nf train --target-dir POOL --name b --epochs 60 --train-mode sameset

# stronger degradation, no curriculum, EMA for flicker
nf train --target-dir POOL --name c --degrade-profile strong \
  --no-degrade-curriculum --ema-decay 0.999
```

```
# per-stage bounds, overriding the profile
nf train --target-dir POOL --name d --degrade-contrast-min 0.3 \
  --degrade-contrast-max 0.4 --degrade-resize-min 0.4
```

```
# whole frame, exact original size (default)
s.infer("photo.jpg")

# tiled; match tile to image_size
s.infer("photo.jpg", tile=256, tile_overlap=64)

# downscale the long edge, aspect preserved, then tile
s.infer("12mp.jpg", max_edge=1024, tile=256)
```